Introduction to Economics

Review for Midterm 1

Blocks 1–4

Pavel Kocourek · Universitat de Barcelona · Monday 16 November 2026

Plan for today

- Midterm 1: the paper and the rules
- Part A: six kinds of question, in pairs
- Part B: the six graph questions
- Three Part B pages, solved on the board
- Thursday: what to bring

Midterm 1 on Thursday: 6 multiple-choice and 6 graph questions

The paper

- Thursday 19 November 2026, 10:05–11:50 (105 minutes)
- Part A: 6 multiple choice, 24 points
- Part B: 6 graph questions on pre-drawn grids, 76 points
- About one point per minute

The rules

- Closed book: pen, paper and any handwritten notes (also photocopied or printed)
- No calculator: every number works out by hand
- No typed or printed text; no formula sheet is provided
- Phones off and away; photo ID on the desk
- Working supports answers; on a graph, the drawing counts as working

Opportunity cost: count what you give up — a sunk cost is never part of it

- Opportunity cost: the best thing you give up + money you spend only because you choose it
- Sunk cost: already paid whatever you choose — leave it out
- Comparative advantage: the lower opportunity cost
- Both gain from trade at any price between the two opportunity costs

Class 2, slide 23 (sunk cost) · class 3, slide 18 (comparative advantage).

Friday's circus show: what does going cost you?

Part A

[4 points]

You paid 28 euros, no refund, for a circus show on Friday evening. Instead you could hand out leaflets: 4 hours at 13 euros an hour. The bus to the show and back costs 6 euros. Nothing else changes. What is your opportunity cost of going to the show?

- **A** 86 euros
- **B** 58 euros
- **C** 52 euros
- **D** 34 euros

Box 1

Answer

A B C D

In pairs, 2 minutes.

Ironing or windows: who should iron, and at what swap?

Part A [4 points]

Two siblings share the housework. Hours each needs:

	one basket of ironing	one window washed
the older	1	¼
the younger	3	½

Both gain from which plan?

- **A** No swap: the older is faster at both.
- **B** The younger irons; swap 1 basket for 5 windows.
- **C** The older irons; swap 1 basket for 7 windows.
- **D** The older irons; swap 1 basket for 5 windows.

Box 2

Answer

A B C D

In pairs, 2½ minutes.

Demand moves for buyers' reasons; elasticity decides what a price rise does to revenue

- The good's own price changes: a move along its demand curve, never a shift
- Demand shifts with incomes, related goods' prices, tastes, expectations, the number of buyers
- Read backwards: price and quantity move the same way $\rightarrow$ demand moved; opposite ways $\rightarrow$ supply moved
- Price elasticity of demand $=$ |% change in quantity $\div$ % change in price|, each % of the midpoint
- Below 1 (inelastic): a higher price raises revenue; above 1 (elastic): it lowers revenue

Inferior good: bought less when incomes rise (class 5, slide 14) · midpoint method: class 7, slides 7–8.

Incomes rise: what happens in the market for canned soup?

Part A

[4 points]

Incomes in a city rise. Canned soup is an inferior good there. What happens in the market for canned soup?

- **A** Demand shifts right.
- **B** Demand shifts left; the lower price then shifts it back right.
- **C** Demand shifts left; the price and the quantity fall.
- **D** Supply shifts left: shops sell less soup.

Box 3

Answer

A B C D

In pairs, 1½ minutes.

A more expensive bowl of ramen: elastic or inelastic, and revenue?

Part A

[4 points]

A ramen bar raises the price of a bowl from 9 to 13 euros. Bowls sold a day fall from 120 to 100. Which statement is correct for this price range?

- **A** Demand is inelastic, and total revenue rises.
- **B** Demand is inelastic, and total revenue falls.
- **C** Demand is elastic, and total revenue rises.
- **D** Demand is elastic, and total revenue falls.

Box 4

Answer

A B C D

In pairs, 2½ minutes.

A monopoly sells while marginal revenue beats marginal cost; a Nash equilibrium is a pair of best replies

- Marginal revenue (MR): what one more unit adds to revenue — the price falls for all
- A monopoly sells every unit whose MR is above its marginal cost (MC)
- Its price: read on the demand row at that quantity
- After a cost rise: the same rule with the new MC
- Nash equilibrium: each player's choice is its best reply to the other's

Monopoly: the only seller, no close substitute (class 12, slide 20) · MR (class 13, slide 4) · Nash equilibrium (class 13, slide 12).

The region's only zip line: which price maximises its profit?

Part A [4 points]

An adventure park runs the region's only zip line. Each extra ride costs it 17 euros (it has no fixed cost). The table gives the highest price at which it can sell each number of rides a day (one price for all riders). Which price maximises its profit?

Rides a day	100	200	300	400	500	600
Price (euros a ride)	35	33	31	29	27	25

(A) 25 euros (B) 19 euros

(C) 17 euros (D) 27 euros

Then: each extra ride costs 8 euros more. The new price?

Box 5

Answer

(A) (B) (C) (D)

In pairs, 2½ minutes.

Two kebab shops, 8 euros or 6: where do they end up?

Part A [4 points]

A small town's only two kebab shops, Brasa and Flama, each charge 8 or 6 euros a kebab, at the same time and without talking. Profits, thousand euros a month:

Brasa's first		Flama: 8 euros	Flama: 6 euros
Brasa	8 euros	Brasa 12, Flama 9	Brasa 4, Flama 12
	6 euros	Brasa 16, Flama 2	Brasa 9, Flama 5

Which outcome is the Nash equilibrium?

(A) Brasa 8, Flama 8 (B) Brasa 8, Flama 6

(C) Brasa 6, Flama 6 (D) Brasa 6, Flama 8

Then: Brasa promises 8 and keeps it. What does Flama charge? What does Brasa earn?

Box 6

Answer

(A) (B) (C) (D)

In pairs, 2 minutes.

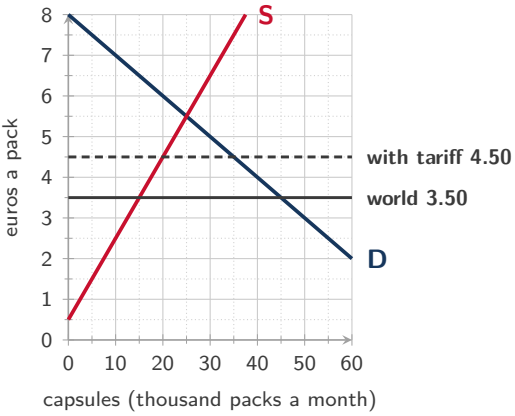
Part B: six graph questions, one move each

B1	Supply and demand from a table	plot, shift, E_1 and E_2, midpoint elasticity
B2	A tax	the tax line; quantity, two prices; CS, PS, T, DWL
B3	A price control	supplied, demanded, shortage or surplus; DWL
B4	Trade and a tariff	world price; production, consumption, imports; T, DWL
B5	An externality	SC or social value; efficient quantity; DWL; corrective tax
B6	A price-taking firm	MC and ATC from TC; best quantity; profit; lowest AVC

Now: B4 and B5 in pairs. After the break: B1, B2, B6 on the board.

D: demand · S: supply · E_1, E_2: the equilibria · CS: consumer surplus · PS: producer surplus · T: tax (or tariff) revenue · DWL: deadweight loss · SC: social cost · TC: total cost · ATC: average total cost (TC ÷ quantity) · AVC: average variable cost.

A tariff on coffee capsules: what is collected, what is lost?



Coffee capsules (packs, thousand a month): a country's demand D and its producers' supply S. It imports at the world price of 3.50 euros a pack; then it puts a tariff of 1 euro on every imported pack.

With the tariff: the imports (thousand packs), T and DWL (thousand euros)?

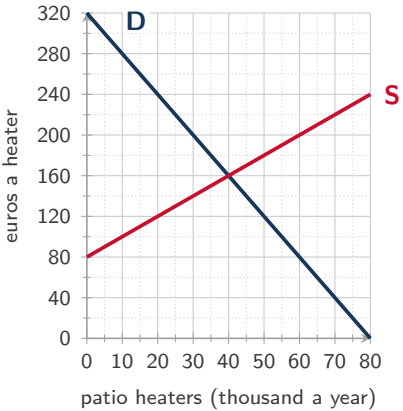
Box 7

Imports	T	DWL

In pairs, 3 minutes.

T: tariff revenue = tariff × imports with the tariff · DWL: deadweight loss.

Patio heaters harm others: how many would be efficient?



Patio heaters (thousand a year): demand D and supply S. The gas each heater burns does 60 euros of harm to other people.

Box 8

Market quantity	Efficient quantity
DWL (thousand euros)	Corrective tax (euros a heater)

In pairs, 2½ minutes.

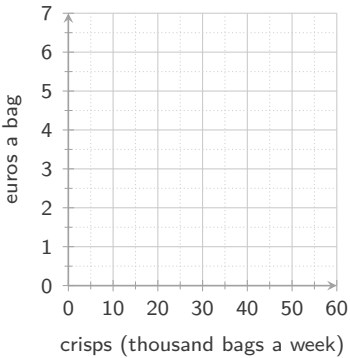
SC: social cost = sellers' cost + the harm to others, at every quantity · DWL: deadweight loss.

Break

Ten-minute break

Crisps: two curves from a table, then a shift — five minutes in pairs

The market for crisps (thousand bags a week). The table gives the quantities demanded and supplied at three prices; both curves are straight lines. [12 points; drawing 6; boxes as marked]



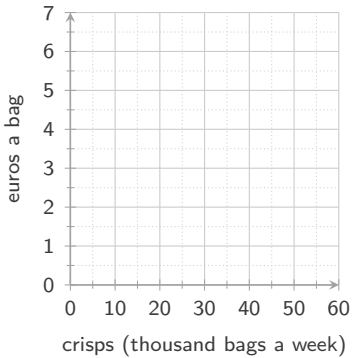
Price (euros a bag)	2	3	4
Quantity demanded (thousand a week)	40	30	20
Quantity supplied (thousand a week)	10	20	30

(a) Draw the demand curve and the supply curve through the table's points, label them D and S, and mark the equilibrium E_1.

(b) The price of frying oil, an input used to make crisps, rises. At every price, sellers now offer 20 thousand fewer bags a week than before. Draw the new supply curve, label it S_2, and mark the new equilibrium E_2.

	Price (euros) [1.5]	Quantity (thousand) [1.5]
Box 9 (a) Before: E_1		
Box 10 (b) After: E_2		

The crisps: the price elasticity of demand from E_1 to E_2 — two minutes in pairs



[2 points]

(c) Demand did not move, so E_1 and E_2 are two points on the demand curve. Compute the price elasticity of demand between them by the midpoint method, and write it as a positive number (a fraction or a decimal).

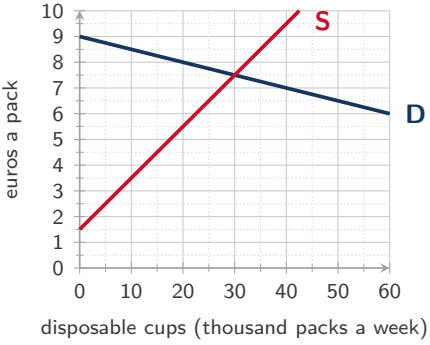
Elasticity [2]

Box 11

Price elasticity of demand: |% change in quantity ÷ % change in price|, each % of the midpoint.

A tax on disposable cups: the line and the prices — three minutes in pairs

The grid shows the market for disposable cups (packs, thousand a week): demand D and supply S. The state puts a tax of 2.50 euros on every pack sold; the sellers pay it to the state.



[5 points; drawing 2; boxes as marked]

(a) Draw the supply curve with the tax: at every quantity 2.50 euros above S. Label it S + tax.

(b) With the tax:

Box 12

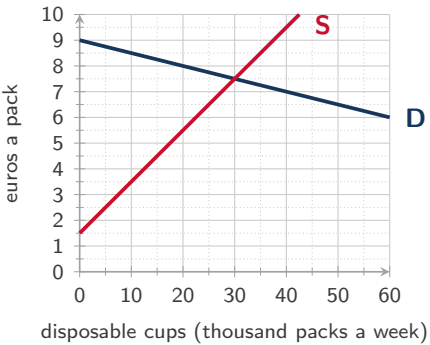
Quantity sold (thousand) [1]

Buyers pay (euros a pack) [1]

Sellers keep (euros a pack) [1]

Draw on the grid and label every line you draw; numbers in the boxes.

The tax's areas as numbers and letters — and who bears more — three minutes in pairs



[9 points; letters 2; boxes as marked]

(c) With the tax, in thousand euros a week: CS, PS, T and DWL. Write each one's letters inside its area.

Box 13

CS [1.5]	PS [1.5]	T [1.5]	DWL [1.5]

(d) Who bears more of the tax? Circle one. [1]

Box 14 Buyers Sellers

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. A triangle is ½ × base × height.

Carnations by the tray: the cost table — three minutes in pairs

A farm grows carnations and sells them by the tray at the market price of 20 euros a tray; it is a price taker. The table gives its total cost (TC) for 0 to 5 trays a day.

(a) Complete the table: the marginal cost (MC) of each tray (the extra cost of that tray) and the average total cost (ATC) at 2, 3 and 4 trays. [5.5 points; table cells as marked]

Box 15

Trays a day	0	1	2	3	4	5
Total cost, TC (euros)	20	30	36	45	60	85
MC of that tray (euros) [0.5]	–					
ATC (euros a tray) [1]	–	–				–

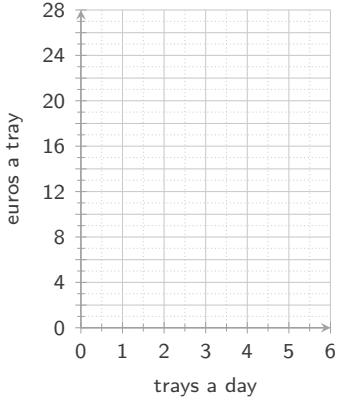
Box 16 What if the farm's rent rises by 10 euros a day?

MC of the 3rd tray

ATC at 4 trays

The farm on the grid: MC, ATC and profit — four minutes in pairs

The same farm, at the market price of 20 euros a tray. Its costs (euros):



[10.5 points; drawing 4.5; boxes as marked]

trays a day	0	1	2	3	4	5
TC	20	30	36	45	60	85
MC						
ATC						

(b) Draw MC as steps: over each tray, a horizontal line at that tray's MC. Mark the ATC at 2, 3 and 4 trays with dots and join them. Draw the price as a line labelled P.

(c) Best quantity (trays a day) [2] Profit (euros a day) [2]

Box 17

Shade the profit as a rectangle; write profit inside it.

(d) At 12 euros a tray: produce or shut down in the short run? Circle one; the reason in one line. [2]

Box 18 (Produce) (Shut down)

MC: marginal cost · ATC: average total cost · AVC: average variable cost, (TC − fixed cost) ÷ trays.

Before Thursday 19 November

- **Midterm 1**: Thursday 19 November, 10:05–11:50, in class (Blocks 1–4)
- Bring: a pen and your photo ID; handwritten notes if you want. No calculator.
- **Class 15**, Monday 23 November: gross domestic product (GDP) and the three ways to measure it
- Read before it: Mankiw & Taylor, Ch. 20, pp. 436–444 (sections 20.2, 20.3)
- Office hours by appointment: e-mail me



Good luck on Thursday!

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